



Ministry of Communications
and Information Technology

DIGITAL EGYPT PIONEERS INITIATIVE · AI & DATA SCIENCE TRACK



Stroke Risk Prediction System

An End-to-End Machine Learning Pipeline, REST API, and Web Application

DEPI Round 4 Graduation Project · Healthcare Predictive Analytics Project

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The Problem

Stroke is one of the leading causes of death and long-term disability worldwide. Early identification of at-risk patients enables preventive intervention.

This project builds a predictive system that estimates a patient's stroke risk from routine demographic, clinical, and lifestyle data — and exposes it through a REST API and web app, not left as a notebook exercise.

~5%

of patients in the dataset had a stroke — a severe class imbalance that shapes every modeling decision in this project.



Objectives

1

Risk Prediction

Train and rigorously compare classifiers to estimate stroke probability.

2

Imbalance-Aware Evaluation

Prioritize ROC-AUC, PR-AUC, recall & F1 over raw accuracy; tune the decision threshold.

3

Leak-Free Pipeline

One scikit-learn Pipeline for preprocessing + model, fit only on the training split.

4

Usable Deployment

FastAPI + Streamlit so a non-technical user gets a real-time risk estimate.

5

Verifiable Testing

Structured functional test cases against the live, running system.

Five Milestones, One Pipeline

The project follows the DEPI Round 4 five-milestone structure for the Healthcare Predictive Analytics Project.

01

**Data Collection &
Preprocessing**

Complete

02

**Advanced Analysis &
Feature Engineering**

Complete

03

**Model Development &
Optimization**

Complete — 6 models

04

**MLOps, Deployment
& Monitoring**

API + Web app live locally

05

**Final Documentation
& Presentation**

This report + live demo

Team & Roles

Each member owned an end-to-end stage of the pipeline — every contribution is demonstrable at the final presentation.



Muuhmd Salah

Team Leader

Project management, requirements & final documentation



Goerge Emil

ML Engineer

Model development & optimization (Milestone 3)



Omar Ahmed

Data Engineer

Data collection, cleaning & preprocessing (Milestone 1)



Toqa Hamed

Data Analyst

Exploratory data analysis & visualization (Milestone 2)



Mostafa Mohamed

Backend / MLOps Engineer

API, deployment & monitoring (Milestone 4)



Nermine Samir

QA & Presentation Lead

Testing, validation & final presentation (Milestone 5)

Dataset Overview & Data Quality

Kaggle Stroke Prediction Dataset — loaded, validated, and cleaned in the EDA notebook (Milestone 1).



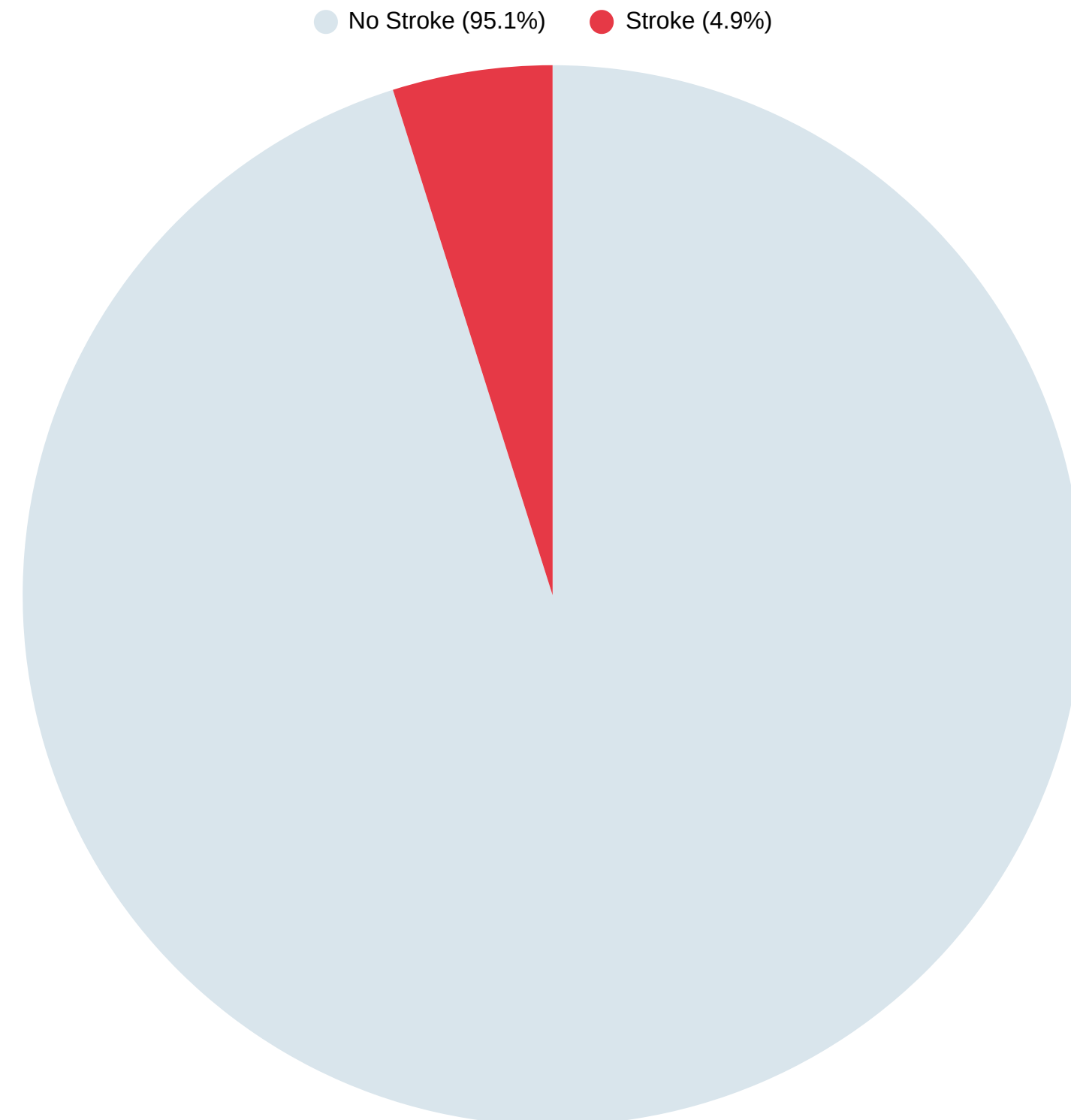
Demographic	gender · age · ever_married · Residence_type
Clinical	hypertension · heart_disease · avg_glucose_level · bmi
Lifestyle	work_type · smoking_status

Two categories are notably rare and flagged for cautious interpretation:

gender = "Other" — only 1 record **work_type = "Never_worked"** — only 22 records (0.4%)

Severe Class Imbalance

Only 249 of 5,110 patients (4.87%) experienced a stroke — this single fact drives the entire evaluation strategy.



A model predicting "no stroke" for everyone would already score

~95% accuracy

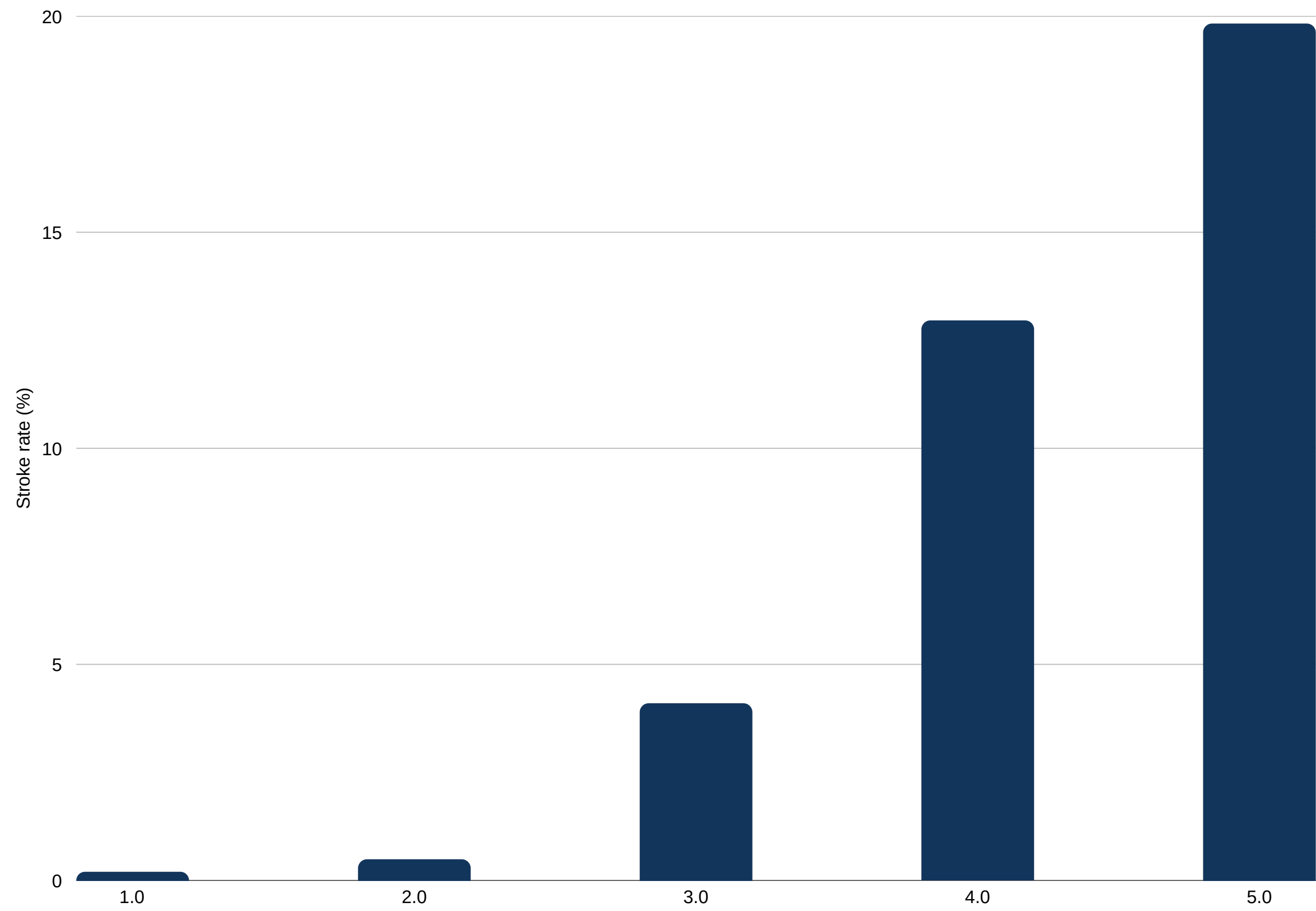
while being clinically useless.

Why the team prioritized other metrics:

- ✓ **Recall** catch as many true stroke cases as possible
- ✓ **F1-score** balance precision and recall
- ✓ **ROC-AUC** overall ranking quality
- ✓ **PR-AUC** performance on the rare positive class

Stroke Risk Rises Sharply With Age

Bivariate analysis (Milestone 2): age is the strongest single risk-related feature in the dataset.



r = 0.245

correlation between age and stroke — the strongest of any numeric feature.

Other correlations with stroke

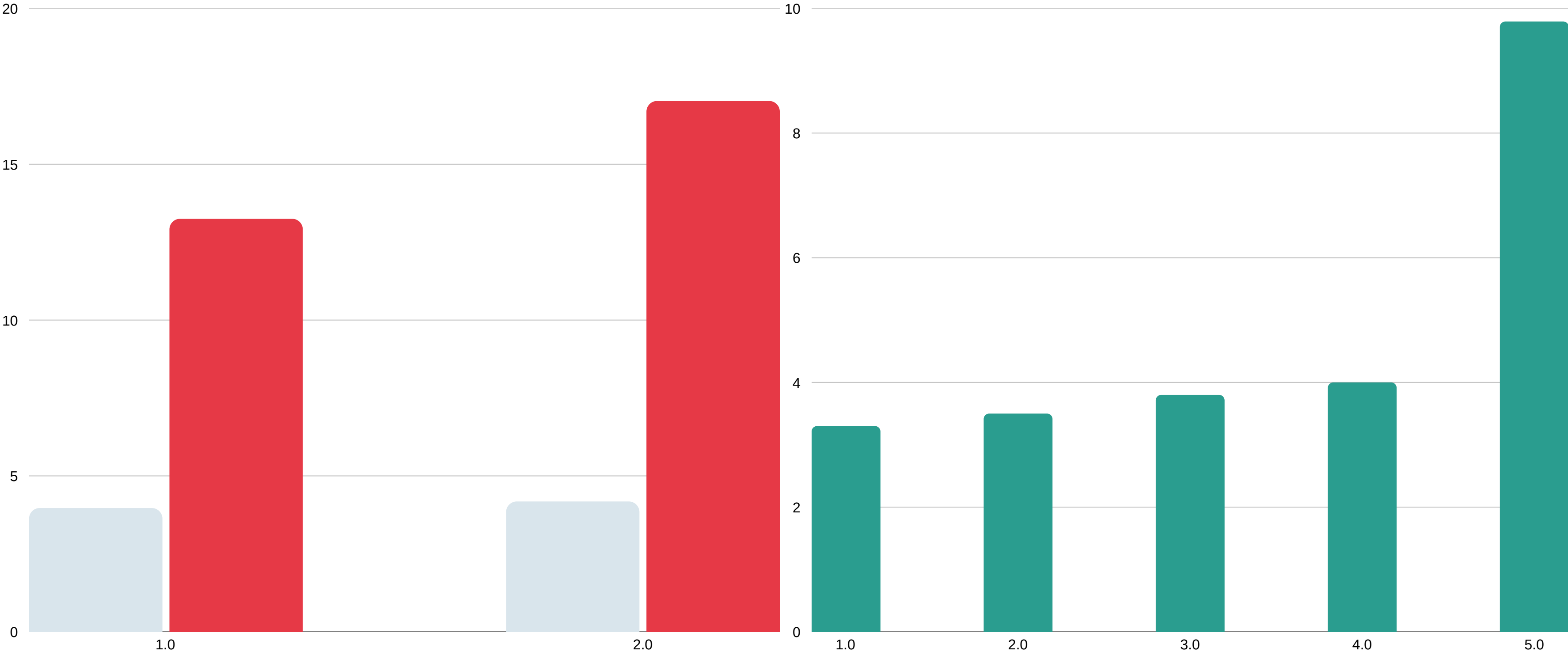
Heart disease	0.135
Avg. glucose level	0.132
Hypertension	0.128
BMI	0.042 (weak)

BMI is only weakly correlated with stroke directly, despite moderate correlation with age (r = 0.333).

Comorbidities Roughly Triple Stroke Risk

Hypertension, heart disease, and elevated glucose all show clear separation between stroke and non-stroke patients.

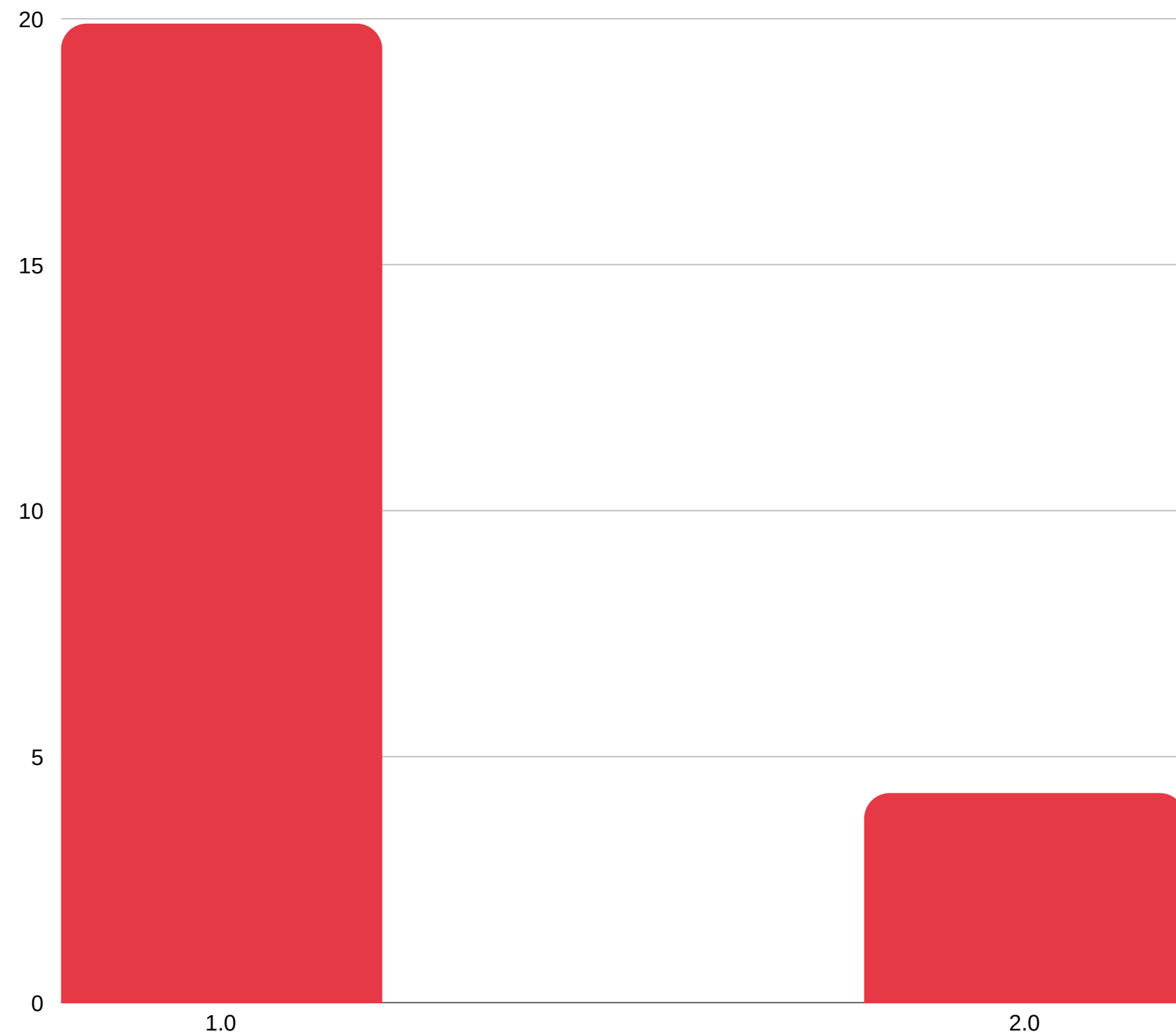
No Yes



The glucose–stroke relationship is concentrated at the high end: flat 3.3%–4.0% for the bottom four quintiles, then a jump to 9.79% in the top quintile.

Missing BMI Is Not Random — It's Informative

Only bmi has missing values: 201 rows (3.93% of the dataset).



$\approx 4.7\times$

gap in stroke rate between patients with missing vs. recorded BMI.

Decision (Milestone 1 preprocessing)

- Median imputation for bmi — robust to outliers
- Add a bmi_missing binary indicator so the model can still use the missingness signal
- Merge the single gender = "Other" row into "Unknown" to avoid an unstable one-hot column

A Leak-Free Preprocessing & Split Strategy



Numeric features (6)

age · hypertension · heart_disease · avg_glucose_level · bmi · bmi_missing — standardized with StandardScaler

Categorical features (5)

gender · ever_married · work_type · Residence_type · smoking_status — one-hot encoded, unseen categories safely ignored

Stratified 60 / 20 / 20 split

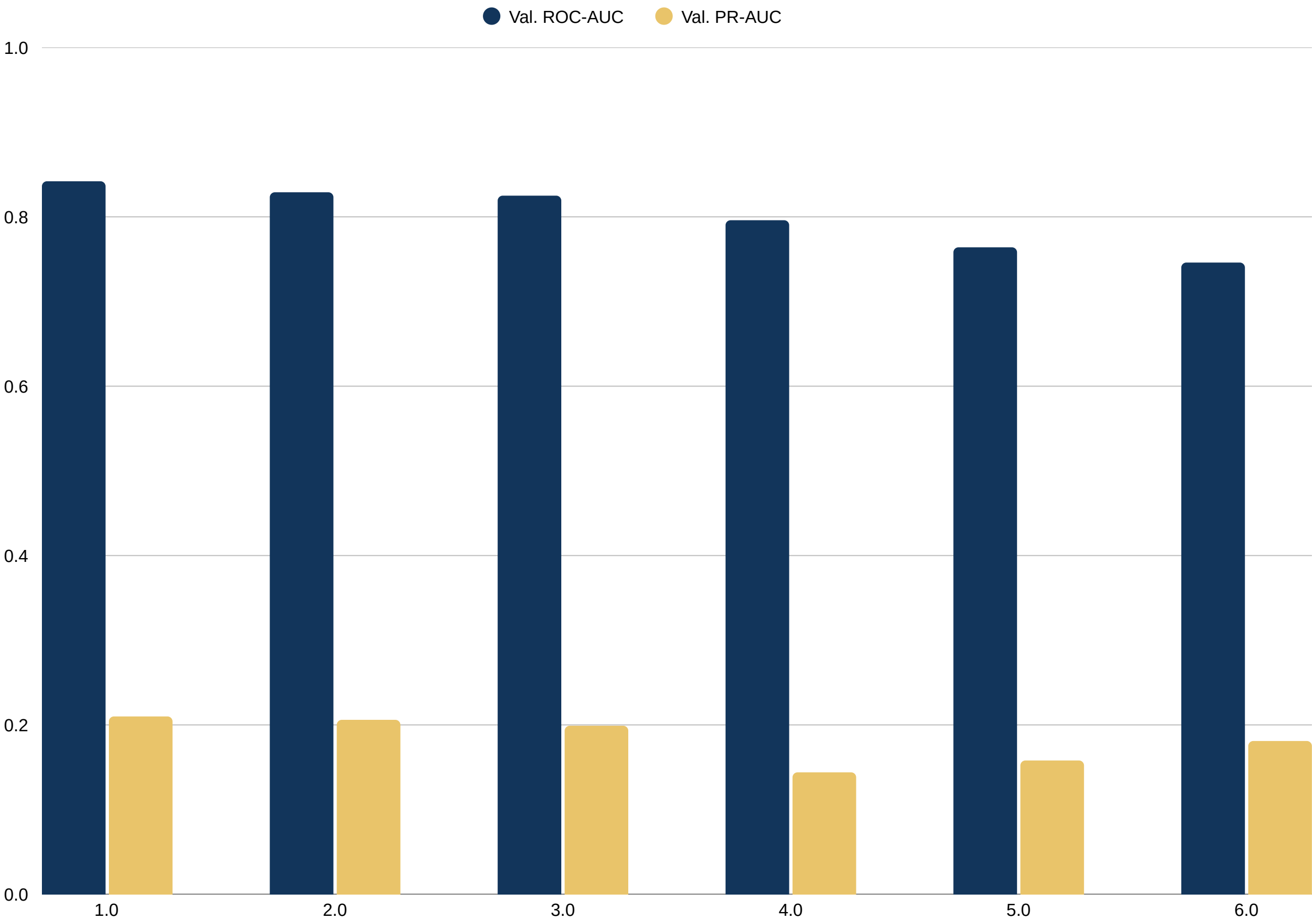
Split	Purpose	Rows	Stroke=1
Train	Fit pipeline & candidate models	3,066	149
Validation	Compare 6 models, tune threshold	1,022	50
Test	One-time final performance	1,022	50

Why three-way split?

The decision threshold is tuned only on the validation set, so the test set reports a single, honest, previously-unseen estimate of performance — no test-set leakage.

Six Models, Compared on Imbalance-Aware Metrics

Validation-set results. Ranked by ROC-AUC and PR-AUC, not accuracy.



Winner

Logistic Regression

Val. ROC-AUC

0.842

Val. PR-AUC

0.210

Val. Recall

0.34

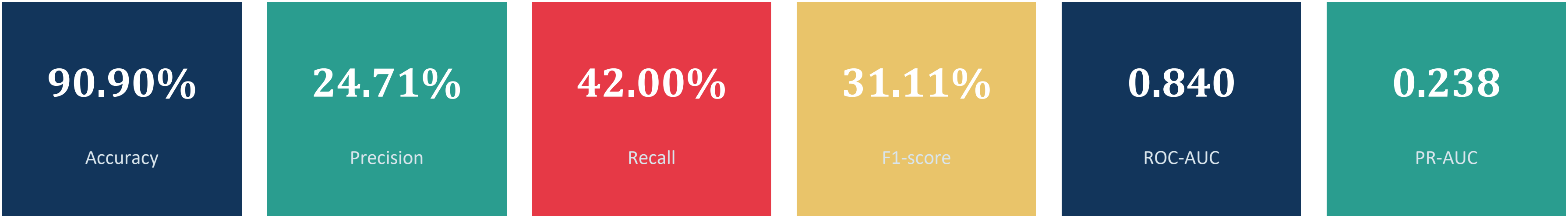
Threshold*

0.803

Selected for leading ROC-AUC & PR-AUC — plus full interpretability, valuable for a healthcare screening tool.

Final Model Performance on the Held-Out Test Set

Logistic Regression, tuned threshold = 0.8028, evaluated once on 1,022 unseen patients (50 with a stroke).



Confusion matrix at the operating threshold

	Pred: No Stroke	Pred: Stroke
Actual: No Stroke	908	64
Actual: Stroke	29	21

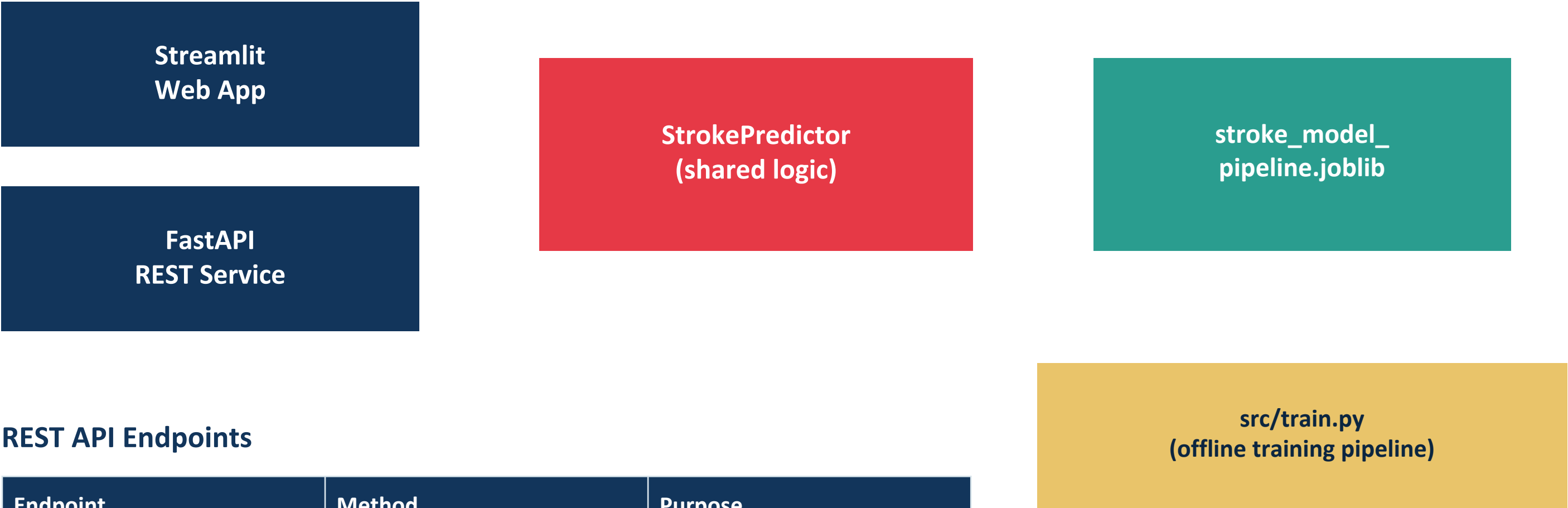
What this means

The model correctly flagged **21 of 50** true stroke cases (42% recall), while producing **64 false positives** among 972 non-stroke patients.

This reflects the precision/recall trade-off on a rare-event problem: catching more true positives requires accepting more false alarms. Consistency between validation ROC-AUC (0.842) and test ROC-AUC (0.840) shows the model is not overfit to the tuning split.

From Notebook to Running Service

Two interfaces share one StrokePredictor class and one serialized pipeline — no train/serve skew.



REST API Endpoints

Endpoint	Method	Purpose
/health	GET	Model load & readiness status
/predict	POST	Score one patient record
/predict-batch	POST	Score multiple records at once

Pydantic v2 validates every request — field types, ranges (age 0–120), and exact trained categories — before it reaches the model.

"The Streamlit web app, backed by the live FastAPI service"

Stroke Prediction

Gender	Work type
Female	Govt_job
Age	Residence type
73	Urban
Hypertension	Average glucose level
No	119.00
Heart disease	BMI
Yes	28.00
Ever married	Smoking status
Yes	never smoked

Predict

Stroke probability

75.04%

No high-risk flag. Threshold: 0.803

Educational project output. Not medical advice.

Stroke Prediction

Gender	Work type
Female	Private
Age	Residence type
45	Urban
Hypertension	Average glucose level
No	100.00
Heart disease	BMI
No	28.00
Ever married	Smoking status
Yes	never smoked

Predict

Stroke probability

19.05%

No high-risk flag. Threshold: 0.803

Educational project output. Not medical advice.

Stroke Prediction

Gender	Work type
Male	Govt_job
Age	Residence type
80	Urban
Hypertension	Average glucose level
No	120.00
Heart disease	BMI
Yes	28.00
Ever married	Smoking status
Yes	formerly smoked

Predict

Stroke probability

84.15%

High risk detected. Threshold: 0.803

Educational project output. Not medical advice.

Tested Against the Live, Running System

Eight realistic patient profiles were submitted to the live /predict endpoint, spanning the full risk spectrum.



Test Profile	Age	Result	Risk
Elderly + hypertension + heart disease + high glucose	78	92.68%	High
Senior smoker, high glucose, BMI not recorded	68	95.90%	High
Elderly, well-controlled (no comorbidities)	81	72.70%	Medium
Middle-aged, obese, high glucose	50	64.20%	Medium
Young adult, no risk factors	24	4.21%	Low
Child (work_type = children)	9	3.91%	Low

All 8 cases passed

Probability ranking matched clinical expectation; the missing-BMI case processed without error, confirming the bmi_missing path works in the live deployment.

Input validation (3 invalid requests)

- ✗ gender = "Other" → 422 rejected
- ✗ age = 200 → 422 rejected
- ✗ missing field → 422 rejected

Honest finding

The 81-year-old with no comorbidities and normal glucose still scored 72.70% — below the high-risk threshold, but far from negligible. Age has an outsized coefficient in the model, so very elderly patients score elevated risk almost regardless of other factors.

Limitations & Future Work



Modest precision

24.7% test precision — ~3 of 4 flagged patients won't go on to have a stroke; the operating point should ultimately be set with clinical input.



Age dominates the score

Elderly patients score elevated risk even without other factors — expected from a linear model, but must be communicated clearly.



Small positive class

Only 249 stroke cases limit how confidently test-set metrics generalize; more labeled data would tighten these estimates.



No resampling explored yet

SMOTE was scoped but not yet applied or compared against the current class-weighting approach.



Tree-based models underperformed

Random Forest, Decision Tree, KNN all trailed Logistic Regression; hyperparameter tuning was not yet applied.



Local-only deployment

Cloud hosting, authentication, and monitoring/retraining automation remain for a production-grade Milestone 4.

A Complete, Working Pipeline

Cleaned & explored data → six-model comparison → tuned threshold → deployed API + web app → verified with functional tests.

0.840

Test ROC-AUC

42%

True Strokes Caught

5/5

Milestones Delivered

The selected Logistic Regression model correctly identifies 42% of true stroke cases at its tuned threshold, with every trade-off documented transparently. Every one of the five DEPI Round 4 milestones has a corresponding, verifiable artifact — and every contribution is traceable to a team member.

Next: cloud deployment · request logging & drift monitoring · resampling experiments · hyperparameter tuning

Thank you

The Future of AI and Mental Health: Collaboration, Innovation, and Responsible Development

The future of AI in mental health is promising, but it requires collaboration, innovation, and responsible development. We must ensure ethical practices, prioritize human interaction, and work towards a brighter future for mental health care.

THANK YOU

